

CSCI2750: Assignment 4

Due: 11:59pm, April 19, 2026; Submit via Gradescope.

INSTRUCTIONS:

1. Answer all questions.
2. Please submit online through **Gradescope** and **assign pages to each question**.
3. By submitting this assignment, you are assumed to have read and understood the course assessment policy, which you can find on Blackboard. You are also assumed to have read and understood CUHK's guideline on academic honesty and on the use of AI tools.
4. All problems are expected to be solved by using the contents in this course. If you have to bring in results out of this scope, prior consent from the instructor shall be sought and a proper citation has to be provided. Exceptions are basic mathematical results in basic calculus and linear algebra.
5. You may discuss with your peers and tutors on these problems. But you must use your own wording and understanding to write up your solution. Similar expressions, equation, sentences, etc., may invite our investigation to turn your submission into an actionable disciplinary case.
6. You may present your work by scans of handwritten pages, or by any tablet-writing applications. However, please keep your submission legible and readable (otherwise we can't grade you...). Also, keep your submission in a single PDF file for the graders'/tutors' convenience.

Problem 1. (20%) Consider the following data points in $\mathbb{R}^3$:

$$\begin{aligned} \mathbf{x}^{(1)} &= (-3, 2, 0), & \mathbf{x}^{(2)} &= (-2, -2, 1), & \mathbf{x}^{(3)} &= (-1, -1, -1), \\ \mathbf{x}^{(4)} &= (0, 0, 1), & \mathbf{x}^{(5)} &= (1, -2, -1), & \mathbf{x}^{(6)} &= (2, 1, 2), \\ \mathbf{x}^{(7)} &= (3, -1, 0). \end{aligned}$$

Suppose that points $\mathbf{x}^{(1)}, \mathbf{x}^{(6)}, \mathbf{x}^{(7)}$ are positive data, and the remaining points are negative. Find a decision tree with 5 nodes assuming that the GINI index is used to measure impurity. Report the corresponding training error.

Problem 2. (20%) Given the following data points:

$$\mathbf{a} = (0, 1, 2, 1), \quad \mathbf{b} = (2, 3, 4, 2), \quad \mathbf{c} = (-2, -1, 1, 3), \quad \mathbf{d} = (1, 0, 1, 0), \quad \mathbf{e} = (3, 4, 5, 3).$$

Let's say, the Manhattan distance is used to measure data similarity. First, construct the dissimilarity matrix $\mathbf{D}$. Then, perform hierarchical clustering on the dataset using the single-linkage method. Sketch the resultant dendrogram.

Hint: For this problem, I don't mind if you use MATLAB or other numerical tools to evaluate $\mathcal{D}$. Nonetheless, try to calculate a few values to ensure you know how to do the computation. (Sorry, no hint here. But it's the only way I had in mind to catch your attention here ☺)

Problem 3. (20%) Consider a one-dimensional dataset:

$$\mathcal{D} = \{x^{(i)} \mid x^{(i)} = 2^i, i = 1, \dots, m\}$$

for a (finite) positive integer m . We will perform clustering by the K -centers algorithm.

- (a) Suppose $m = 6$, and the initial center is $z_1 = x^{(1)}$. Let $K = 3$. Perform K -centers clustering on $\mathcal{D}$. Report i) all the center points z_k 's, ii) the resultant cluster groupings $\mathcal{Z}_k$'s, and iii) the radius $r(\mathcal{Z})$ of the clustering result.
- (b) Compute the SSE and BSS measurement of the clustering result in (a).
- (c) Suppose m is no longer fixed at 6. Proof:

Theorem. If the 2-centers algorithm is ran on $\mathcal{D}$, then the resultant clusters have $r(\mathcal{Z}) \leq 2^{m-1}$.

Problem 4. (20%) Below lists the items included in different travellers' packing lists.

Traveller	Items in the Packing List
Aaron	Phone Charger, Passport
Bianca	Water Bottle, Passport, Power Bank, Phone Charger
Chloe	Water Bottle, Phone Charger, Power Bank
Daniel	Water Bottle, Phone Charger
Evelyn	Passport, Water Bottle, Phone Charger
Felix	Power Bank, Water Bottle, Phone Charger
Hannah	Water Bottle, Passport

Suppose $\text{minsup} = 0.7$ and $\text{minconf} = 0.8$.

- (a) Write down the set of unique items $\mathcal{I}$ appearing in the dataset.
- (b) Use the Apriori method to mine all frequent itemsets.
- (c) List out all the strong association rules.
- (d) Which item would you recommend Aaron to add to the packing list? Briefly explain.
- (e) Which item would you recommend Hannah to add to the packing list? Briefly explain.

Problem 5. (20%) In this problem we explore how to implement truncated SVD for image compression. We consider gray-scale images for (a)–(c). Color images are considered in (d).

- (a) Following the instructions stated below in the appendix, study and understand the MATLAB codes used to test the truncated SVD compression method.

- (b) Select an image by yourself. Implement the SVD compression method on it. Determine what is the largest value of k that would make your compression meaningful. Choose a suitable value of k and discuss on how did you obtain this k . Visualize the reconstructed image $\bar{\mathbf{X}}$ and the original image $\mathbf{X}$ as in the format of the Figure in the Appendix.
- (c) Visualize the results of $\bar{\mathbf{X}}$ output by your SVD algorithm with the compression rate of 10%, 20%, 40% and 80%. Here, compression rate refers to the ratio of

$$\text{compression rate} = \frac{\text{storage}(\bar{\mathbf{X}})}{\text{storage}(\mathbf{X})}.$$

List the corresponding k 's and comment on which compression rate do you recommend to be used for your image.

- (d) Modify the above codes/operations to do SVD compression on RGB image, i.e., perform SVD compression on each color channel. Observe the result like what I did in p.28–29 in Lecture 12.

Appendix: Implementation of SVD Compression

The MATLAB implementation of compression using the singular value decomposition is straightforward. It only reads the following few lines:

```
X = imread('testing.png');
X = rgb2gray(X);           % convert to gray-scale
X = double(X);
X = X./max(X(:));          % normalization
% where the built-in SVD algorithm is used
[U, S, V] = svd(X);
```

The resultant SVD matrices $\mathbf{U}$, $\mathbf{\Sigma}$ and $\mathbf{V}$ are returned at the argument $\mathbf{U}$, $\mathbf{S}$, $\mathbf{V}$, respectively. Next, we read the corresponding *smaller (or compressed)* matrices

$$\bar{\mathbf{U}} = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_k \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}, \quad \bar{\mathbf{\Sigma}} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_k \end{bmatrix}, \quad \bar{\mathbf{V}}^T = \begin{bmatrix} \leftarrow & \mathbf{v}_1^T & \rightarrow \\ \leftarrow & \mathbf{v}_2^T & \rightarrow \\ & \vdots & \\ \leftarrow & \mathbf{v}_k^T & \rightarrow \end{bmatrix}$$

by the following command lines

```
U_new = U(:, 1:k);
S_new = diag(S); S_new = diag(S_new(1:k));
V_new = V(:, 1:k);
```

and finally reassemble $\bar{\mathbf{X}} = \bar{\mathbf{U}}\bar{\mathbf{\Sigma}}\bar{\mathbf{V}}^T$ by the command of `X_new = U_new*S_new*V_new.'`. You may visualize the results of the approximation by using:

```
figure();  
subplot(2,1,1); imshow(X);  
subplot(2,1,2); imshow(X_new);
```



Figure: Visualization of X (left) and its approximation $\bar{X}$ (right).